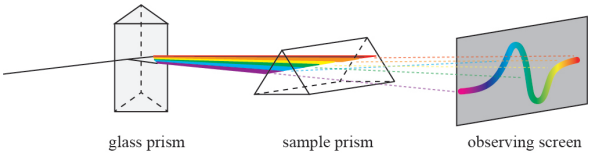
Exercises Part I

Semiclassical Quantum Optics

EXERCISE 1. CLASSICAL MODEL OF RESONANT OPTICAL RESPONSE

In 1902, Robert Williams Wood expanded on August Kundt's method of crossed prisms to map out the refractive index of sodium vapor near its famous "D" absorption line. In a classic crossed-prisms experiment, white light passes through a vertical glass prism, which spreads the colors out horizontally into a spectrum (like a standard rainbow). This horizontally dispersed light then passes through a horizontal prism filled with sodium vapor.



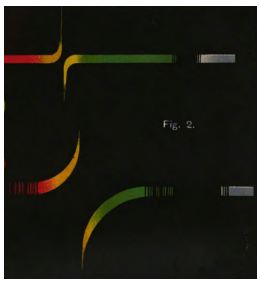


FIG. 1. Crossed-prisms experiment and the observed spectrum.

(a) Model the optical susceptibility using the Lorentz model. Determine the refractive index and absorption coefficient assuming low absorption.

(b) Comment and explain the observations of the Fig. 1.

(c) Make a model for the deflection observed on the screen using Snell's law, and comment on how to determine the free parameters of the Lorentz model from these observations.

(d) Can you observe slow light (i.e. ultra-low light velocities) in this experiment?

EXERCISE 2. THE MOLLWO-IVEY RELATION

An F-center consists of an electron trapped at a missing anion in an ionic crystal. We may approximate the electron as a particle of mass m_e confined to a three-dimensional cubic

infinite potential well of side length a:

$$V(x,y,z) = \begin{cases} 0, & 0 < x, y, z < a, \\ +\infty, & \text{otherwise.} \end{cases} \tag{1}$$

Assume that the confinement length a is proportional to the crystal lattice spacing, and neglect electron spin.

(a) Solve the time-independent Schrödinger equation by separation of variables. Apply the boundary conditions and normalize the wavefunctions. Show that the energy eigenvalues are

$$E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2 m_e a^2} \left(n_x^2 + n_y^2 + n_z^2 \right), \qquad n_x, n_y, n_z = 1, 2, \ldots . \tag{2}$$

(b) Identify the ground state and the first excited energy level. Calculate their energies and determine the degeneracy of the first excited level.

(c) For light polarized along x, the dipole interaction is $\hat{V}(t) = eE(t)\hat{x}$, where $e > 0$. Using the product structure of the wavefunctions, determine which states of the first excited level can be reached from the ground state by an electric-dipole transition. Verify that the relevant matrix element of $\hat{x}$ is nonzero.

(d) Let ν_{abs} denote the frequency of the lowest allowed absorption transition. Use

$$h\nu_{\text{abs}} = E_{\text{exc}} - E_{\text{g}} \tag{3}$$

to derive the dependence of ν_{abs} on a. Show that

$$\nu_{\text{abs}} a^2 = \text{constant}, \tag{4}$$

and calculate the constant within this model. This inverse-square scaling is the particle-in-a-box prediction for the *Mollwo–Ivey relation*.

(e) For two crystals described by the same model, show that

$$\ln\!\left(\frac{a_2}{a_1}\right) = -\frac{1}{2} \ln\!\left(\frac{\nu_{\text{abs},2}}{\nu_{\text{abs},1}}\right). \tag{5}$$

Does increasing the lattice spacing produce a red shift or a blue shift? Calculate the percentage change in the absorption frequency when the confinement length increases by 10%. Finally, discuss two limitations of modelling a real colour centre as an infinite cubic well.

EXERCISE 3. RABI OSCILLATIONS, POPULATION LOSS, AND THE DENSITY MATRIX

Consider a two-level atom driven by a classical field $E(t) = E_0 \cos \omega t$, with Hamiltonian

$$\hat{H}(t) = \hbar\omega_0 |2\rangle\langle 2| - \hbar\Omega \cos \omega t \left(|2\rangle\langle 1| + |1\rangle\langle 2| \right), \tag{6}$$

where $\Omega = dE_0/\hbar$ is real and $\Delta = \omega_0 - \omega$. The atom is initially excited: $|\psi(0)\rangle = |2\rangle$.



FIG. 2. Level schemes for Exercise 3. (Original diagram from *Rand, Nonlinear and Quantum Optics* using the density matrix Exercise 3.10 and Exercise 3.11.)

(a) Starting from the Schrödinger equation, transform to the interaction picture with respect to $\hat{H}_0 = \hbar\omega_0 |2\rangle\langle 2|$. Identify the terms oscillating at $\omega_0 - \omega$ and $\omega_0 + \omega$, and state the conditions for the rotating-wave approximation (RWA).

After applying the RWA, introduce rotating-frame amplitudes through

$$C_1 = e^{-i\Delta t/2} b_1, \qquad C_2 = e^{i\Delta t/2} b_2. \tag{7}$$

Derive the resulting time-independent Hamiltonian and solve for the populations $P_1(t)$ and $P_2(t)$. Identify the generalized Rabi frequency, discuss the resonant limit, and verify that $P_1 + P_2 = 1$.

(b) Now suppose that population leaves each level for states outside the two-level manifold [Fig. 2(a)], at rates Γ_1 and Γ_2, respectively. The interaction-picture amplitudes then satisfy

$$\begin{aligned} \dot{C}_2 &= -\frac{\Gamma_2}{2} C_2 + \frac{i\Omega}{2} e^{i\Delta t} C_1, \\ \dot{C}_1 &= -\frac{\Gamma_1}{2} C_1 + \frac{i\Omega}{2} e^{-i\Delta t} C_2. \end{aligned} \tag{8}$$

Use the same rotating-frame transformation to solve these equations for $C_2(0) = 1$ and $C_1(0) = 0$, and obtain $|C_1(t)|^2$ and $|C_2(t)|^2$. Show that

$$\frac{d}{dt}\left(|C_1|^2 + |C_2|^2\right) = -\Gamma_1|C_1|^2 - \Gamma_2|C_2|^2. \tag{9}$$

Interpret the missing probability and recover part (a) when both decay rates vanish.

(c) For a closed two-level manifold, spontaneous emission returns population from $|2\rangle$ to $|1\rangle$ [Fig. 2(b)]. Consider the proposed amplitude equations

$$\begin{aligned} \dot{C}_2 &= -\frac{\Gamma}{2}C_2 + \frac{i\Omega}{2}e^{i\Delta t}C_1, \\ \dot{C}_1 &= +\frac{\Gamma}{2}C_2 + \frac{i\Omega}{2}e^{-i\Delta t}C_2. \end{aligned} \tag{10}$$

Calculate the time derivative of $|C_1|^2 + |C_2|^2$ and show that these equations do not generally preserve normalization. Evaluate that derivative at $t = 0$ for the specified initial state.

Explain why transferring population at rate $\Gamma|C_2|^2$ is not equivalent to adding an amplitude source $\Gamma C_2/2$. Write instead the spontaneous-emission contribution to $\dot{\rho}_{11}$, $\dot{\rho}_{22}$, and $\dot{\rho}_{12}$, and verify that the density-matrix description preserves $\rho_{11} + \rho_{22} = 1$.

EXERCISE 4. PURE STATES, MIXTURES, AND ATOMIC COHERENCE

Consider a two-level atom with basis $\{|1\rangle, |2\rangle\}$ and transition dipole operator

$$\hat{p} = d\Big(|1\rangle\langle 2| + |2\rangle\langle 1|\Big),$$

where d is real. The density operator of a pure state is $\rho = |\psi\rangle\langle\psi|$; for a statistical ensemble, $\rho = \sum_a P_a |\psi_a\rangle\langle\psi_a|$.

(a) Write the density matrices for the ground state $|1\rangle$ and excited state $|2\rangle$. Explain the physical meaning of diagonal and off-diagonal entries, and show that $\rho_{11} + \rho_{22} = 1$.

(b) For $|\psi\rangle = C_1|1\rangle + C_2|2\rangle$, with $|C_1|^2 + |C_2|^2 = 1$, calculate ρ_{12} and ρ_{21}. Show how these elements contain the relative phase but make no contribution to the populations.

(c) Compare an equal coherent superposition $|+\rangle = (|1\rangle + |2\rangle)/\sqrt{2}$ with an equal incoherent mixture of $|1\rangle$ and $|2\rangle$. Write both density matrices. Determine their populations,

coherences, $\mathrm{Tr}(\rho^2)$, and mean electric dipoles. Explain why one density matrix cannot be obtained from the other by a unitary basis change.

(d) Let $|\psi\rangle = C_1 |1\rangle + e^{i\phi} C_2 |2\rangle$, where ϕ varies independently and uniformly over $[0, 2\pi)$ across an ensemble. Obtain the ensemble-averaged density matrix and the average dipole. Explain the distinction between a coherent superposition for each atom and coherence of the ensemble.

EXERCISE 5. DRIVEN TWO-LEVEL ATOM WITH DECAY, DEPHASING, AND INCOHERENT PUMPING

A closed two-level transition $|1\rangle \leftrightarrow |2\rangle$ of frequency ω_0 is driven by $E(t) = E_0 \cos(\omega t)$. Define

$$\Omega = \frac{dE_0}{\hbar} > 0, \qquad \Delta = \omega_0 - \omega, \qquad \sigma_+ = |2\rangle\langle 1|, \quad \sigma_- = |1\rangle\langle 2|, \quad \sigma_z = |2\rangle\langle 2| - |1\rangle\langle 1|.$$

Spontaneous emission transfers $|2\rangle \to |1\rangle$ at rate Γ, pure dephasing occurs at rate γ_ϕ, and an incoherent pump transfers $|1\rangle \to |2\rangle$ at rate R. All rates are nonnegative. Assume $\omega \approx \omega_0$ and that the dipole moment is real.

(2.1) Start from

$$H(t) = \hbar\omega_0 |2\rangle\langle 2| - \hbar\Omega \cos(\omega t)(\sigma_+ + \sigma_-). \tag{11}$$

Moving to the the interaction picture, identify the slowly rotating and counter-rotating terms; state conditions for the rotating-wave approximation (RWA). Then set

$$U(t) = |1\rangle\langle 1| + e^{-i\omega t} |2\rangle\langle 2|, \qquad \rho_{\mathrm{rot}} = U^\dagger \rho U,$$

and derive why the effective hamiltonian in this frame follows

$$H_{\mathrm{rot}} = U^\dagger H_{\mathrm{RWA}} U - i\hbar U^\dagger \dot{U} = \hbar \begin{pmatrix} 0 & -\Omega/2 \\ -\Omega/2 & \Delta \end{pmatrix}. \tag{12}$$

(2.2) Use the Lindblad master equation

$$\dot{\rho}_{\mathrm{rot}} = -\frac{i}{\hbar}[H_{\mathrm{rot}}, \rho_{\mathrm{rot}}] + \Gamma \mathcal{D}[\sigma_-]\rho_{\mathrm{rot}} + \frac{\gamma_\phi}{2}\mathcal{D}[\sigma_z]\rho_{\mathrm{rot}} + R\mathcal{D}[\sigma_+]\rho_{\mathrm{rot}}, \tag{13}$$

where

$$\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}.$$

Justify why the dissipators retain this form in the rotating frame. Derive equations for ρ_{22} and $\widetilde{\rho}_{21} = (\rho_{\rm rot})_{21}$, use $\rho_{11} = 1 - \rho_{22}$, and identify the population and coherence relaxation rates. Check trace preservation.

(2.3) Set the derivatives to zero and solve for $\rho_{11}^{\rm ss}$, $\rho_{22}^{\rm ss}$, and $\widetilde{\rho}_{21}^{\rm ss}$ as functions of Ω, Δ, Γ, γ_ϕ, and R (take $\Gamma + R > 0$). For an atomic number density $\mathcal{N}$, use

$$E(t) = \mathrm{Re}[E_0 e^{-i\omega t}], \qquad P(t) = \mathrm{Re}[\widetilde{P} e^{-i\omega t}], \qquad \widetilde{P} = 2\mathcal{N} d\, \widetilde{\rho}_{21}^{\rm ss}, \qquad \chi(\omega, E_0) = \frac{2\mathcal{N} d^2}{\epsilon_0 \hbar \Omega}\, \widetilde{\rho}_{21}^{\rm ss}. \tag{14}$$

Derive the complex susceptibility and its weak-drive limit. Expand the formula and distinguish the linear susceptibility from the intensity-dependent effective susceptibility.

(2.4) Discuss how increasing R changes (i) the no-drive population difference, (ii) the coherence linewidth, and (iii) absorption and dispersion. Identify the cases $R < \Gamma$, $R = \Gamma$, and $R > \Gamma$ in the weak-drive limit. Can the pump produce gain? Does a stronger pump necessarily increase the magnitude of the susceptibility?

EXERCISE 6. THREE-LEVEL Λ SYSTEM: DARK STATE AND EIT

Consider an atom with long-lived lower states $|1\rangle$, $|2\rangle$ and excited state $|3\rangle$. A weak probe drives $|1\rangle \leftrightarrow |3\rangle$ with complex Rabi frequency Ω_p; a control field drives $|2\rangle \leftrightarrow |3\rangle$ with Ω_c. Use $\Delta_p = \omega_{31} - \omega_p$, $\Delta_c = \omega_{32} - \omega_c$, and $\delta = \Delta_p - \Delta_c$. In a suitable rotating frame and after the RWA, take

$$\frac{H_{\rm rot}}{\hbar} = \Delta_p \left|3\right\rangle\left\langle 3\right| + \delta \left|2\right\rangle\left\langle 2\right| - \frac{1}{2}\left(\Omega_p \left|3\right\rangle\left\langle 1\right| + \Omega_c \left|3\right\rangle\left\langle 2\right| + \text{h.c.}\right). \tag{15}$$

(a) Write the 3×3 Hamiltonian matrix in the ordered basis $\{|1\rangle, |2\rangle, |3\rangle\}$. Explain the physical meanings of one-photon and two-photon detuning.

(b) At $\delta = 0$, construct a normalized state $|D\rangle$ that has no excited-state amplitude and satisfies $H_{\rm rot}|D\rangle = 0$. Explain why this is called a dark state.

(c) Assume a weak probe, $|\Omega_p| \ll |\Omega_c|$, with $\rho_{11}^{(0)} \simeq 1$. Optical coherence $\widetilde{\rho}_{31}$ decays at rate γ_{31}, and lower-state coherence $\widetilde{\rho}_{21}$ at γ_{21}. Derive the first-order stationary equations for these two coherences and solve for $\widetilde{\rho}_{31}^{(1)}$.

(d) Given the probe susceptibility

$$\chi_p = \frac{2\mathcal{N}d_{31}^2}{\epsilon_0 \hbar \Omega_p}\, \widetilde{\rho}_{31}^{(1)},$$

calculate χ_p. Discuss the limit $\delta = 0$, $\gamma_{21} \to 0$, and explain how ground-state dephasing limits transparency.